Row Echelon Form and Reduced Row Echelon Form of a Matrix

Row Echelon Form:

A matrix is in **row echelon form**, when it satisfies the following conditions.

- The first non-zero element in each row, called the **leading entry**, is 1.

  (Note: Some authors don't require that the leading coefficient is a 1. It could be any number.)
- Each **leading entry** is in a **column** to the **right** of the **leading entry** in the **previous row**.
- **Rows** with **all zero elements**, if any, are below **rows** having a **non-zero element**.

Each of the matrices shown below are examples of matrices in **row echelon form**.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Example:

Which of the following matrices is in **row echelon form**?

a) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$

b) $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

c) $C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$

d) $D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$

Solution:

The correct answer is **option (b)**, i.e., **Matrix B is in row echelon form**, since it satisfies all of the requirements for a row echelon matrix. The other matrices fall short.

- The leading entry in Row 1 of matrix **A** is to the right of the leading entry in Row 2, which is inconsistent with definition of a row echelon matrix.
- In matrix **C**, the leading entries in Rows 2 and 3 are in the same column, which is not allowed.
- In matrix **D**, the row with all zeros (Row 2) comes before a row with a non-zero entry.

Reduced Row Echelon Form

A matrix is in **reduced row echelon form**, when it satisfies the following conditions.

- The matrix satisfies conditions for a row echelon form.
- The leading entry in each row is the only non-zero entry in its column.

Each of the matrices shown below are examples of matrices in ***reduced*** **row echelon form**.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Example:

Which of the following matrices is in **reduced row echelon form**?

a) $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b) $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

c) $C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

d) **All of above**

e) **None of the above**

Solution:

The correct answer is **option (d)**, since each matrix satisfies all of the requirements for a reduced row echelon matrix.

- The first non-zero element in each row, called the **leading entry**, is 1.
- Each leading entry is in a column to the right of the leading entry in the previous row.
- Rows with all zero elements, if any, are below rows having a non-zero element.
- The leading entry in each row is the only non-zero entry in its column.

Question:

For each of the following matrices, determine if it is reduced row-echelon form.

a) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **(Answer: No)**

b) $\begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}$ **(Answer: Yes)**

c) $\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ **(Answer: Yes)**

d) $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **(Answer: No)**

e) $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ **(Answer: Yes)**

Augmented Matrices and Elementary Row Operations

As the number of equations and unknowns in a linear system increases so does the complexity of the algebra involved in finding solutions, we write system of equations in the form of matrices and solve it by

- **Row Echelon Form** (Gauss Elimination Method), or
- **Reduced Row Echelon Form** (Gauss Jorden Method).

These methods are very useful to find the solutions of system of linear equations.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \qquad\qquad\qquad \cdots\cdot \\ \qquad\qquad\qquad \cdots\cdot \\ \qquad\qquad\qquad \cdots\cdot \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

The ***augmented matrix*** for the system of linear equations is

$$\begin{bmatrix} a_{11} & a_{12} & \cdots\cdot & a_{1n} & : b_1 \\ a_{21} & a_{22} & \cdots\cdot & a_{2n} & : b_2 \\ \vdots & & \ddots & & \vdots \\ a_{m1} & a_{m2} & \cdots\cdot & a_{mn} & : b_m \end{bmatrix}$$

The basic method for solving a linear system is to perform appropriate algebraic operations on the system that do not alter the solution set and that produce a succession of increasingly simpler systems, until a point is reached where it can be determined whether the system is consistent, and if so, what its solutions are. Typically, the algebraic operations are as follows:

1. Multiply an equation by a non-zero constant.

2. Interchange two equations.

3. Add a constant (scalar) times one equation to another.

Since the rows (horizontal lines) of an augmented matrix correspond to the equations in the associated system, these three operations correspond to the following operations on the rows of the augmented matrix:

1. Multiply a row by a non-zero constant.

2. Interchange two rows.

3. Add a constant (scalar) times one row to another.

These are called **elementary row operations** on a matrix.

Gauss Elimination Method

Example 1: Use elementary row operations to solve the linear system using Gauss Elimination Method.

$$\begin{cases} x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - 3x_3 = 1 \\ 3x_1 + 6x_2 - 5x_3 = 0 \end{cases}$$

Solution: The Augmented matrix is

$$\begin{bmatrix} 1 & 1 & 2 & :9 \\ 2 & 4 & -3 & :1 \\ 3 & 6 & -5 & :0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & :9 \\ 0 & 2 & -7 & :17 \\ 3 & 6 & -5 & :0 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & :9 \\ 0 & 2 & -7 & :17 \\ 0 & 3 & -11 & :-27 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 3 & -11 & : & -27 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 \times \left(\frac{1}{2}\right)$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & : & -\frac{3}{2} \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 0 & 1 & : & 3 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 \times (-2)$$

Echelon Form

$$\begin{cases} x_3 = 3 \\ x_2 - \frac{7}{2}x_3 = -\frac{17}{2} \\ x_1 + x_2 + 2x_3 = 9 \end{cases}$$

Putting $\boldsymbol{x_3 = 3}$ in equation (2), we get $\boldsymbol{x_2 = 2}$

Using $\boldsymbol{x_3 = 3, x_2 = 2}$ in equation (3), we get $\boldsymbol{x_1 = 1}$

So $(\boldsymbol{x_1, x_2, x_3}) = (\boldsymbol{1, 2, 3})$ is solution of above system.

Gauss Jordan Method

Finding the solution of linear system by converting the augmented matrix into reduced echelon form is called Gauss Jorden Method.

Example: Use elementary row operations to solve the linear system using **Gauss Jorden Method**.

$$\begin{cases} \boldsymbol{x_1 + x_2 + 2x_3 = 9} \\ \boldsymbol{2x_1 + 4x_2 - 3x_3 = 1} \\ \boldsymbol{3x_1 + 6x_2 - 5x_3 = 0} \end{cases}$$

Solution: The Augmented matrix is

$$\begin{bmatrix} 1 & 1 & 2 & : 9 \\ 2 & 4 & -3 & : 1 \\ 3 & 6 & -5 & : 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & : 9 \\ 0 & 2 & -7 & : 17 \\ 3 & 6 & -5 & : 0 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & : 9 \\ 0 & 2 & -7 & : 17 \\ 0 & 3 & -11 & : -27 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 3 & -11 & : & -27 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 \times \left(\frac{1}{2}\right)$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & : & -\frac{3}{2} \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 1 & \frac{-7}{2} & : & -\frac{17}{2} \\ 0 & 0 & 1 & : & 3 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 \times (-2)$$

Echelon Form

Now, We will convert it into reduced echelon form,

$$\begin{bmatrix} 1 & 1 & 2 & : 9 \\ 0 & 1 & 0 & : 2 \\ 0 & 0 & 1 & : 3 \end{bmatrix} \text{-------------} R_2 + \frac{7}{2}R_3$$

$$\begin{bmatrix} 1 & 1 & 0 & : 3 \\ 0 & 1 & 0 & : 2 \\ 0 & 0 & 1 & : 3 \end{bmatrix} \text{-------------} R_1 - 2R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & : 1 \\ 0 & 1 & 0 & : 2 \\ 0 & 0 & 1 & : 3 \end{bmatrix} \text{-------------} R_1 - R_2$$

Reduced Row Echelon form

$$x_1 = 1,\ x_2 = 2,\ x_3 = 3$$

So $(\boldsymbol{x_1, x_2, x_3}) = (\mathbf{1, 2, 3})$ is solution of above system.

Example 2: Solve the linear system of equation:

$$\begin{cases} \boldsymbol{x+z+2w=6} & ----\rightarrow (\mathbf{1}) \\ \boldsymbol{y-2z=-3} & ------\rightarrow (\mathbf{2}) \\ \boldsymbol{x+2y-z=-2} & -----\rightarrow (\mathbf{3}) \\ \boldsymbol{2x+y+3z-2w=0} & ----\rightarrow (\mathbf{4}) \end{cases}$$

Solution:

Using (1) and (4), we get

$$\boldsymbol{3x+y+4z=6} -----\rightarrow (\mathbf{5})$$

Using equation (3) and (5)

$$\boldsymbol{5y-7z=-12} -----\rightarrow (\mathbf{6})$$

Using equation (2) and (6)

$$\boldsymbol{-3z=-3} \quad \textbf{or} \quad \boldsymbol{z=1}$$

Put $\boldsymbol{z=1}$ in equation (6) implies

$$\boldsymbol{y=-1}$$

Putting value of $\boldsymbol{y}$ and $\boldsymbol{z}$ in equation (3), we get

$$\boldsymbol{x=1}$$

Finally, using value of $\boldsymbol{x, y}$ and $\boldsymbol{z}$ in equation (1) we have

$$\boldsymbol{w=2}$$

So $(\boldsymbol{x,y,z,w}) = (\mathbf{1,-1,1,2})$ is solution of above system.

Example: Solve the linear system of equations using **Gauss-Jordan method**:

$$\begin{cases} \boldsymbol{x+z+2w=6} & ---\rightarrow (\mathbf{1}) \\ \boldsymbol{y-2z=-3} & ------\rightarrow (\mathbf{2}) \\ \boldsymbol{x+2y-z=-2} & -----\rightarrow (\mathbf{3}) \\ \boldsymbol{2x+y+3z-2w=0} & ----\rightarrow (\mathbf{4}) \end{cases}$$

Solution: The Augmented matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 2 & : & 6 \\ 0 & 1 & -2 & 0 & : & -3 \\ 1 & 2 & -1 & 0 & : & -2 \\ 2 & 1 & 3 & -2 & : & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 & : & 6 \\ 0 & 1 & -2 & 0 & : & -3 \\ 0 & 2 & -2 & -2 & : & -8 \\ 0 & 1 & 1 & -6 & : & -12 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - R_1, \;\; R_4 - 2R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 & : & 6 \\ 0 & 1 & -2 & 0 & : & -3 \\ 0 & 0 & 2 & -2 & : & -2 \\ 0 & 0 & 3 & -6 & : & -9 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 2R_2, \;\; R_4 - R_2$$

$$\begin{bmatrix} 1 & 0 & 1 & 2: & 6 \\ 0 & 1 & -2 & 0: & -3 \\ 0 & 0 & 1 & -1: & -1 \\ 0 & 0 & 3 & -6: & -9 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad \frac{1}{2}R_3$$

$$\begin{bmatrix} 1 & 0 & 1 & 2: & 6 \\ 0 & 1 & 0 & -2: & -5 \\ 0 & 0 & 1 & -1: & -1 \\ 0 & 0 & 0 & -3: & -6 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim R_1 - R_3,\ \ R_2 + 2R_3,\ \ R_4 - 3R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 3: & 7 \\ 0 & 1 & 0 & -2: & -5 \\ 0 & 0 & 1 & -1: & -1 \\ 0 & 0 & 0 & 1: & 2 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad -\frac{1}{3}R_4$$

$$\begin{bmatrix} 1 & 0 & 0 & 0: & 1 \\ 0 & 1 & 0 & 0: & -1 \\ 0 & 0 & 1 & 0: & 1 \\ 0 & 0 & 0 & 1: & 2 \end{bmatrix} \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_1 - 3R_4,\ \ R_2 + 2R_4,\ \ R_4 - 3R_3$$

Thus, $\boldsymbol{x = 1, \quad y = -1, \quad z = 1, \quad w = 2.}$

Hence, $\boldsymbol{(x, y, z, w) = (1, -1, 1, 2)}$

Some Facts About Echelon Forms

There are three facts about row echelon forms and reduced row echelon forms that are important to know but we will not prove:

1. Every matrix has a unique reduced row echelon form; that is, regardless of whether you use Gauss-Jordan elimination or some other sequence of elementary row operations, the same reduced row echelon form will result in the end.

2. Row echelon forms are not unique; that is, different sequences of elementary row operations can result in different row echelon forms.

3. Although row echelon forms are not unique, all row echelon forms of a matrix *A* have the same number of zero rows, and the leading 1's always occur in the same positions in the row echelon forms of A. Those are called the ***pivot positions*** of *A*. A column that contains a pivot position is called a ***pivot column*** of *A*.

Example 4: Use **Gauss-Jordan** elimination to solve the homogeneous linear system

$$\begin{cases} z + w = 0 \\ y + z = 0 \\ x + y = 0 \\ x + w = 0 \end{cases}$$

Hint: Solution is $(x, y, z, w) = (t, -t, t, -t)$.

Solution:

The above given system is converted into coefficient matrix form as

$$\begin{array}{c} \begin{array}{cccc} x & y & z & w \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Interchange Row 1 with row 4, we have

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Now applying row operation as

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} ----------- R_3 - R_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} ----------- R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} ----------- R_4 + R_3$$

We can write as

$$\begin{array}{c} \begin{array}{cccc} x & y & z & w \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Since, all entries (elements) of last row becomes zero. This is called ZERO ROW. Converting this matrix into equations, we have

$$x + w = 0 ------- (1)$$

$$y + z = 0 ------- (2)$$

$$-z - w = 0 -------(3)$$

Now, we have 3 equations and 4 variables, which is called under-determined system. Therefore, in this situation, we have to free one variable.

Let

$$\boldsymbol{x = t\ (Free\ variable)\ ----------\ (4)}$$

Put equation (4) in equation (1), we get

$$\boldsymbol{t + w = 0 \quad \Longrightarrow \quad w = -t}$$

Put $\boldsymbol{w = -t}$ in equation (3), we get

$$\boldsymbol{-z - w = 0 \quad \Longrightarrow \quad -z - (-t) = 0 \quad \Longrightarrow -z + t = 0 \quad \Longrightarrow \ t = z \ \Longrightarrow \quad z = t}$$

Put $\boldsymbol{z = t}$ in equation (2), we get

$$\boldsymbol{y + z = 0 \quad \Longrightarrow \quad y + t = 0 \quad \Longrightarrow \quad y = -t}$$

So,

$$x = t, \qquad y = -t, \qquad z = t, \qquad w = -t$$

Hence,

$$\boldsymbol{(x, y, z, w) = (t, -t, t, -t)}$$

Howard Anton (Exercise 1.2)

Q1. In each part, determine whether the matrix is in row echelon form, reduced row echelon form, both or neither.

$$(a)\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad (b)\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad (c)\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(d)\ \begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 4 \end{bmatrix} \qquad (e)\ \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad (f)\ \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(g)\ \begin{bmatrix} 1 & -7 & 5 & 5 \\ 0 & 1 & 3 & 2 \end{bmatrix}$$

Q3. In each part, suppose that the augmented matrix for a system of linear equations has been reduced by row operations to the given reduced row echelon form. Solve the system.

$$(a)\ \begin{bmatrix} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix} \qquad (b)\ \begin{bmatrix} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

$$(c)\ \begin{bmatrix} 1 & 7 & -2 & 0 & -8 & -3 \\ 0 & 0 & 1 & 1 & 6 & 5 \\ 0 & 0 & 0 & 1 & 3 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad (d)\ \begin{bmatrix} 1 & -3 & 7 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Work to do:

Q5 Solve the linear system of equation by using Gauss elimination or Gauss-Jordan elimination method

$$\begin{cases} x_1 + x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 + 3x_3 = 1 \\ 3x_1 - 7x_2 + 4x_3 = 10 \end{cases}$$

Gauss-Jordan Method

Example 1: Solve the linear system of equation using Gauss-Jordan method

$$\begin{cases} x + 4y - z = 12 \\ 3x + 8y - 2z = 4 \end{cases}$$

Solution. The **augmented matrix** of the given system of linear equation is

$$\begin{bmatrix} 1 & 4 & -1 & :12 \\ 3 & 8 & -2 & :4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & -1 & :12 \\ 0 & -4 & 1 & :-32 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 4 & -1 & :12 \\ 0 & 1 & -\frac{1}{4} & :8 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad -\frac{1}{4}R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & :-20 \\ 0 & 1 & -\frac{1}{4} & :8 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_1 - 4R_2$$

By **back substitution**, we get the equations below

$$1x + 0y + 0z = -20 \quad \Longrightarrow \boldsymbol{x = -20}$$

$$0x + 1y - \frac{1}{4}z = 8 \quad \Longrightarrow \boldsymbol{y - \frac{1}{4}z = 8}$$

Let $\boldsymbol{z = t}$, then

$$\boldsymbol{y = 8 + \frac{1}{4}t}$$

So, the solution of this system is $(\boldsymbol{x, y, z}) = \left(\boldsymbol{-20,\ 8 + \frac{1}{4}t,\ t}\right)$.

Example 2: Solve the following systems of linear equations using Gauss Jordan elimination method:

$$\begin{cases} x + 2y + z = 12 \\ 5x + 2y - 3z = 1 \\ 2x + y - z = 2 \end{cases}$$

Solution: The **augmented matrix** of the given system of linear equation is

$$\begin{bmatrix} 1 & 2 & 1 & :12 \\ 5 & 2 & -3 & :1 \\ 2 & 1 & -1 & :2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & :12 \\ 0 & -8 & -8 & :-59 \\ 2 & 1 & -1 & :2 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 - 5R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & :12 \\ 0 & -8 & -8 & :-59 \\ 0 & -3 & -3 & :-22 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & :12 \\ 0 & 1 & 1 & :\frac{59}{8} \\ 0 & -3 & -3 & :-22 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad -\frac{1}{8}R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & :12 \\ 0 & 1 & 1 & :\frac{59}{8} \\ 0 & 0 & 0 & :\frac{1}{8} \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 + 3R_2$$

In last row, all coefficients of the given variables become **ZERO.** The result becomes

$$\mathbf{0 = \frac{1}{8} \quad \Longrightarrow \quad 0 = \boldsymbol{constant} \quad \Longrightarrow \boldsymbol{No\ solution}}$$

Hence, the above given system **has no solution.**

Example 2: Solve the following systems of linear equations using Gauss Jordan elimination method.

$$\begin{cases} x + 2y + 3z = 6 \\ 2x - 3y + 2z = 14 \\ 3x + y - z = -2 \end{cases}$$

Solution: The **augmented matrix** of the given system of linear equation is

$$\begin{bmatrix} 1 & 2 & 3 & :6 \\ 2 & -3 & 2 & :14 \\ 3 & 1 & -1 & :-2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & :6 \\ 0 & -7 & -4 & :2 \\ 3 & 1 & -1 & :-2 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & :6 \\ 0 & -7 & -4 & :2 \\ 0 & -5 & -10 & :-20 \end{bmatrix} \quad \sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim \quad R_3 - 3R_1$$

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$$\begin{bmatrix} 1 & 0 & 0 & : 1 \\ 0 & 1 & 0 & : -2 \\ 0 & 0 & 1 & : 3 \end{bmatrix}$$

So, the above system has solution $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = (\mathbf{1}, -\mathbf{2}, \mathbf{3})$.

Example 4: Solve the following systems of linear equations using Gauss Jordan elimination method:

$$\begin{cases} x + 2y - 3z = -4 \\ 2x + y - 3z = 4 \end{cases}$$

Solution: Do yourself

The above system has solution $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = (\mathbf{4} + \boldsymbol{t}, \boldsymbol{t} - \mathbf{4}, \boldsymbol{t})$.

Work to do:

Howard Anton (Exercise 1.2)

Question 1: Solve the following systems of linear equations using Gauss Jordan elimination method.

I. $$\begin{cases} 2x + 2y + 4z = 0 \\ w - y - 3z = 0 \\ 2w + 3x + y + z = 0 \\ -2w + x + 3y - 2z = 0 \end{cases}$$

Answer:

$$w = t, \quad x = -t, \quad y = t, \quad z = 0, \quad i.e., \qquad (w, x, y, z) = (t, -t, t, 0)$$

II. $$\begin{cases} x_1 + 3x_2 + x_4 = 0 \\ x_1 + 4x_2 + 2x_3 = 0 \\ -2x_2 - 2x_3 - x_4 = 0 \\ 2x_1 - 4x_2 + x_3 + x_4 = 0 \\ x_1 - 2x_2 - x_3 + x_4 = 0 \end{cases}$$

III. $$\begin{cases} 2x_1 - x_2 + 3x_3 + 4x_4 = 9 \\ x_1 - 2x_3 + 7x_4 = 11 \\ 3x_1 - 3x_2 + x_3 + 5x_4 = 8 \\ 2x_1 + x_2 + 4x_3 + 4x_4 = 10 \end{cases}$$

Answer:

$$x_1 = -1, x_2 = 0, x_3 = 1, x_4 = 2, \qquad \Longrightarrow (x_1, x_2, x_3, x_4) = (-1, 0, 1, 2)$$

<u>Question 2:</u> Determine the values of $\boldsymbol{a}$ for which the system has **no solutions, exactly one solution, or infinite many solutions**.

$$\begin{cases} \boldsymbol{x + 2y - 3z = 4} \\ \boldsymbol{3x - y + 5z = 2} \\ \boldsymbol{4x + y + (a^2 - 14)z = a + 2} \end{cases}$$

Answer:

- If $\boldsymbol{a = 4}$**,** there are **infinitely many solutions**.
- If $\boldsymbol{a = -4}$**,** there is **no solution**.
- If $\boldsymbol{a \neq \pm 4}$**,** there is **exactly one solution** or **unique solution**.